

REVIEW

Application of machine learning for signal recognition in distributed fibre optic acoustic sensing technology

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Abstract

Coherent Rayleigh scattering-based distributed fibre optic sensing technology enables real-time acquisition of vibration and acoustic information along the optical fibres. However, the complexity of monitoring environments often leads to false alarms and missed detections during the process of information source identification with distributed acoustic sensing (DAS). Therefore, it becomes crucial to effectively extract meaningful signal features and perform accurate pattern recognition in the presence of external noise disturbance. The authors provide a comprehensive review of signal feature extraction and pattern recognition techniques applied in DAS technology. After introducing the fundamentals of DAS, specific applications are considered, and the following techniques have been analysed and compared: feature extraction algorithms based on wavelet decomposition, feature extraction schemes utilising other decomposition models, traditional recognition classifiers, and neural network-based recognition classifiers using deep learning. The advantages and limitations of each scheme are discussed, along with their potential applications in various scenarios. The aim is to provide insights into the latest technologies in signal processing and pattern recognition for DAS, fostering further advancements in this field.

KEYWORDS

distributed sensors, object detection, optical engineering computing, optical fibre communication

1 | INTRODUCTION

Based on the principle of phase-sensitive optical time-domain reflectometry (φ -OTDR), distributed acoustic sensing (DAS) is an innovative sensing technology that achieves distributed detection by demodulating the Rayleigh back-scattered light, enabling high-precision perception and localisation of environmental vibration sources and acoustic field information [1]. This technology not only possesses the advantages of conventional distributed fibre optic sensing but also exhibits characteristics of high response speed and sensitivity. Since its proposal by Taylor, et al. in 1993, significant attention was garnered by DAS [2].

The development process of DAS has been divided into three stages: qualitative detection, quantitative detection, and performance enhancement [3]. During the qualitative detection

stage, real-time detection of disturbances along the distributed optical fibre has been carried out using pulse disturbance. This stage has commonly been referred to as distributed vibration sensing (DVS) and has formed the basis of DAS technology. In the quantitative detection stage, external information has been quantitatively extracted by demodulating the phase of Rayleigh back-scattered light. DAS has found widespread applications. The performance enhancement stage has focused on optimising critical parameters of the DAS system, such as spatial resolution and response bandwidth, to improve the system's performance. Throughout the development of DAS technology, researchers have conducted a series of studies to enhance the performance of DAS systems [4]. With the continuous improvement of technical performance, fibre optic sensing technology has experienced rapid development [5] and is currently widely used in various fields such as pipeline leak

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detection [6], waterway safety [7], perimeter security [8], railway safety monitoring [9], and geological exploration [10].

This paper presents a comprehensive review of the research progress in feature extraction schemes and recognition classifiers used in identification systems, building upon the introduction of DAS and incorporating specific applications. Additionally, it offers an outlook on feature extraction and pattern recognition.

2 | DISTRIBUTED OPTIC FIBRE ACOUSTIC SENSING TECHNOLOGY

Distributed optic fibre sensing technology is a real-time, continuous, and distributed monitoring technique, and within this field, φ -OTDR is a significant technical scheme. φ -OTDR is an implement way of DAS. Based on the principle of φ -OTDR, DAS enables real-time monitoring of the external environment by detecting changes in the properties of the sensing optical fibre.

Fibre optic sensing signals exhibit high sensitivity, excellent resolution, wide dynamic range, and long sensing distances. They feature fast response times, good linearity, and stability, while also being able to resist external interference, ensuring high reliability. These characteristics enable precise monitoring of subtle changes within the optical fibre. It was reported that the spatial resolution has been improved from the metre level to the centimetre level, and the sensing distances has exceeded 100 km [5]. A typical schematic diagram is shown in Figure 1.

Efficient and rational feature extraction and pattern recognition techniques can expand the application scope of fibre optic sensing technology and provide more reliable technical support for practical applications. However, during the subsequent identification of various vibration signals, missing or false alarms may occur.

In recent years, with the continuous improvement of signal processing techniques, there has been significant progress in the research into feature extraction and recognition of fibre optic vibration signals. These research efforts can be summarised into the following steps: (1) After preprocessing the acquired signals, appropriate feature extraction schemes are selected to extract time-frequency domain features. (2) Suitable pattern recognition schemes are employed to identify target events. At the same time, to mitigate the disturbance of noise and improve the system's recognition rate, signal denoising is often performed before feature extraction.

3 | FEATURE EXTRACTION SCHEMES

With the continuous development of distributed fibre optic sensing technology, higher accuracy and real-time capabilities are required for the identification of fibre optic sensing signals in specific applications, making the effectiveness of feature extraction increasingly crucial. Feature selection should meet three criteria: first, the selected features should maximise the differences between different types of signals, while minimising

the differences within the same type of signals. Second, the selected features should represent important attributes of the original signals, fully reflecting the characteristics and patterns of the signals. Finally, the selected features should exhibit good stability, meaning that the recognition performance on different signal sets should be relatively stable and less susceptible to disturbance from noise and other factors.

3.1 | Wavelet-based feature extraction scheme

The wavelet-based feature extraction is a widely used approach that involves transforming signals using wavelet analysis, decomposing them into different frequency components, and extracting signal features. In 1992, a neural network structure based on wavelet functions was employed to signal processing and pattern recognition tasks [11]. In 2017, a convolutional neural network (CNN) was employed to classify time-frequency images generated through wavelet transforms [12]. Wavelet analysis was employed to domains such as image processing, audio processing, and data compression [13]. The specific model of the wavelet-based feature extraction scheme is shown in Table 1.

A scheme utilising wavelet packet decomposition for the analysis of four distinct events in practical applications was proposed in 2022 based on the φ -OTDR system [14]. This scheme, according to the signal's frequency characteristics, has employed Dmeyer wavelet basis for a three-level wavelet packet decomposition. The energy spectra of the decomposed typical wavelet packets are shown in Figure 2a, and the average energy distribution of the four types of signals is shown in Figure 2b.

From Figure 2, differences in the energy distribution of signals of different types can be observed, indicating that this scheme effectively represents signal features.

In response to the problem of false alarms in complex environments, a scheme for event recognition that extracts comprehensive time-frequency domain features was proposed in 2019 [15]. In this scheme, the frequency energy vectors of the signal were extracted for feature analysis using a 4th level wavelet packet decomposition. The frequency domain energy characteristics are shown in Figure 3.

From Figure 3, significant differences in the time-frequency domain features of the four types of events can be observed. Experimental results indicated that this scheme achieves an average recognition rate of 98.3% for these four typical intrusion events.

The feature extraction scheme based on wavelet decomposition has advantages in multi-scale analysis, local feature extraction, and robustness to noise. It has been found suitable for processing non-stationary signals and signals with significant frequency variations. However, in practical applications, it exhibits a relatively high computational complexity and requires the selection of appropriate wavelet basis functions and scale parameters based on the actual signal characteristics and application requirements.

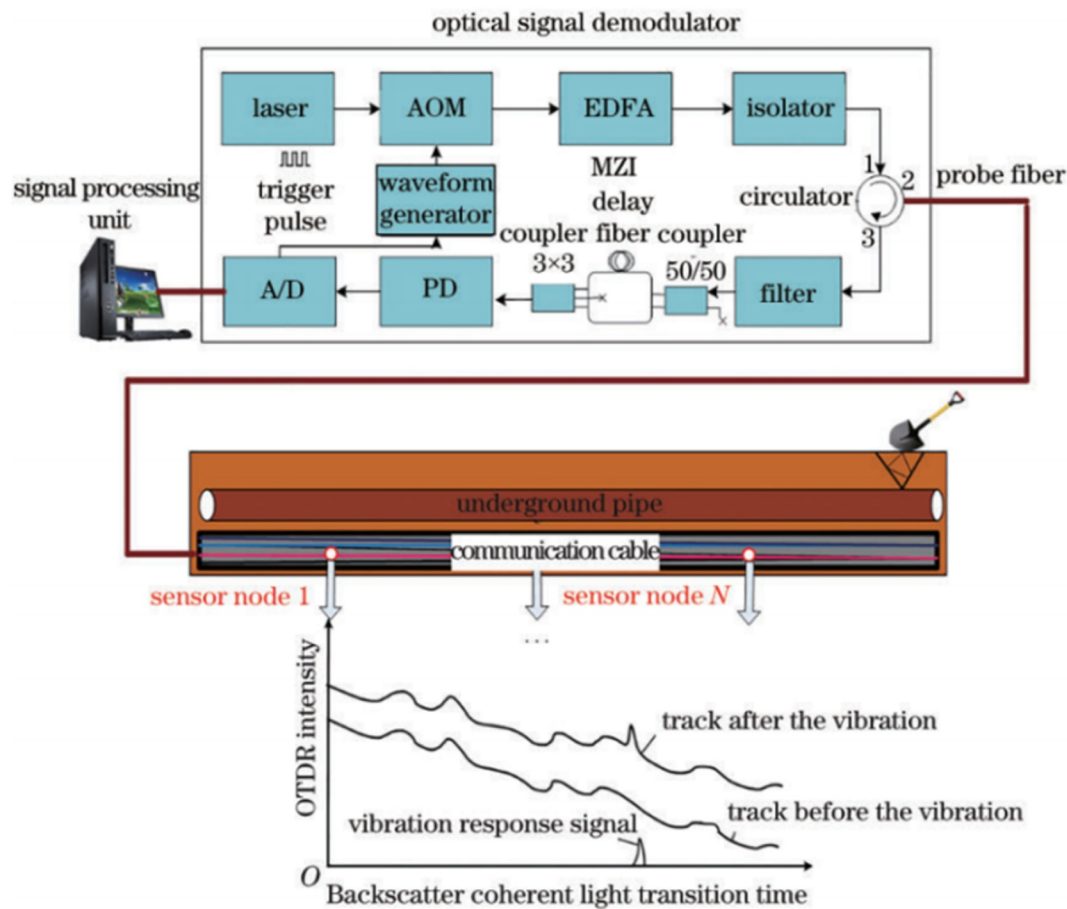


FIGURE 1 Principle of DVS/DAS based on ϕ -OTDR (AOM [Acousto-Optic Modulator]; PD [Photodetector]; EDFA [Erbium-Doped Fibre Amplifier]; A/D [Analogue-to-Digital Converter]) [1]. A/D, analogue-to-digital converter; AOM, acousto-optic modulator; DAS, distributed acoustic sensing; DVS, distributed vibration sensing; EDFA, erbium-doped fibre amplifier; PD, photodetector.

TABLE 1 Feature extraction scheme based on wavelet decomposition.

Feature extraction scheme	Ref.	Specific identification event	Average recognition accuracy (%)
Wavelet-based feature extraction scheme	[14]	Shaking, knocking, walking, and ambient noise	97.0
	[15]	Footsteps of passerby, bicycle rolling, knocking on the fence, and cutting of optical cable	98.3

3.2 | Feature extraction schemes based on other decomposition models

In the field of distributed fibre optic sensing, besides feature extraction schemes based on wavelet decomposition, there are various schemes available for signal analysis. These schemes include Wigner–Ville Distribution (WVD), Spectral Euclidean Distance Method (EDFS), waveform analysis, Fourier transform, empirical mode decomposition (EMD), and linear discriminant analysis (LDA), among others. Each scheme possesses unique feature extraction capabilities and can adapt to different types of signals. The specific models of feature extraction schemes based on other decomposition models are shown in Table 2.

3.2.1 | Feature extraction scheme based on Wigner–Ville distribution

WVD is a signal processing technique based on time–frequency analysis. It calculates the WVD of a signal to extract its time–frequency features and analyse its time–frequency characteristics. The concept of WVD was first introduced in 1932 [24] and later applied to the field of signal processing. This scheme was found suitable for analysing signals with transient and non-stationary characteristics, such as transient speech features in speech signals, fault features in vibration signals, motion features in image sequences etc. [25].

A scheme utilising WVD to extract features from the raw signals of underwater targets was proposed in 2022. The

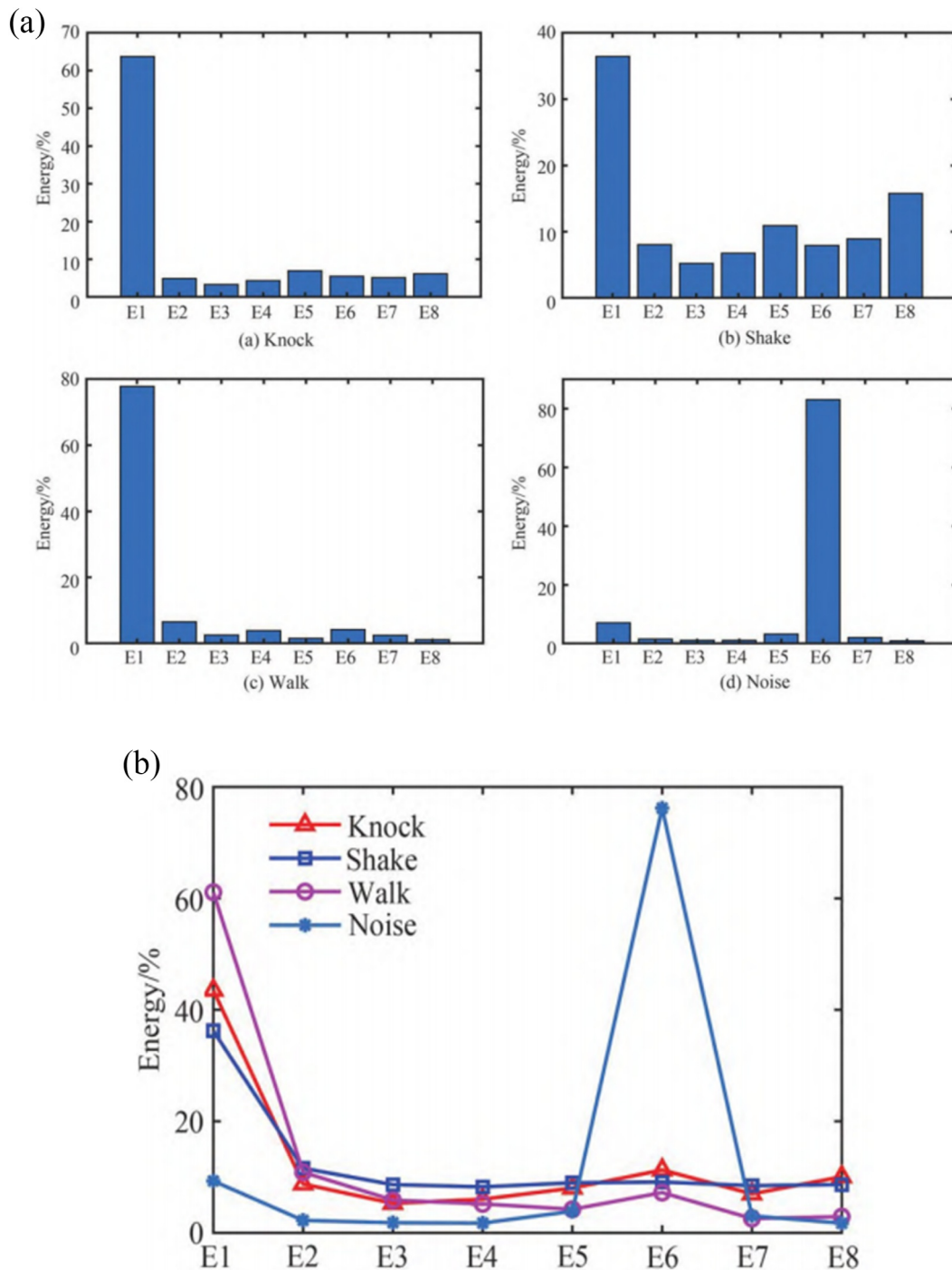


FIGURE 2 (a) Typical energy spectra of three-level wavelet packet decomposition [14]. (b) The average energy distribution of signals for four different events [14].

accuracy of identifying different types of underwater targets has reached 97.8% [16].

In response to the safety monitoring problems around building glass windows, a monitoring scheme based on WVD and extreme gradient boosting trees was proposed in 2020 [17]. By artificially simulating eight types of vibration events, feature vector was extracted using WVD. Experimental results

indicated that the average recognition rate for the eight types of events has reached 93.3%.

WVD has advantages in providing high time–frequency resolution, conducting instantaneous frequency analysis, and adapting to non-stationary signals. However, it has higher computational complexity, is more sensitive to noise, and the features extracted using WVD are typically mathematical time–

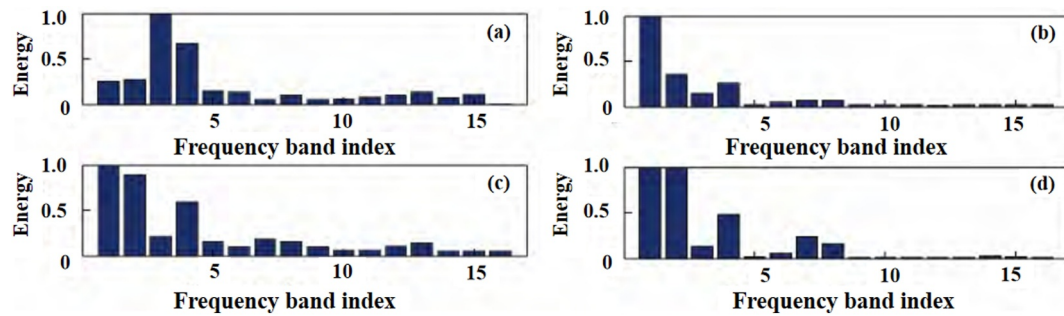


FIGURE 3 Frequency domain energy distribution (a) pedestrian footsteps; (b) bicycle passing; (c) hitting the fence; (d) cutting the fibre optic cable [15].

TABLE 2 Feature extraction schemes based on other decomposition models.

Feature extraction scheme	Ref.	Specific identification event	Average recognition accuracy (%)
Feature extraction scheme based on Wigner–Ville distribution	[16]	Underwater target recognition	97.8
	[17]	Wind blowing, knocking, window opening, watering, hammering, dog barking, aeroplane noise, and undisturbed	93.3
EDFS	[18]	Kick wall, stomp wall, and run in place	
The feature extraction scheme based on waveform characteristics	[19]	Shovel, hoe, jackhammer, and hammer	94.8
The feature extraction scheme based on Fourier transform	[20]	Tap, shake, gust of wind, and rain	97.2
	[21]	Train location	
EMD	[22]	Pipeline leakage monitoring	94.7
Feature extraction schemes based on linear discriminant analysis	[23]	Walking, tapping, and rolling wheel compaction	92.6

Abbreviations: EDFS, spectral Euclidean distance method; EMD, empirical mode decomposition.

frequency distribution functions, which may have relatively limited physical interpretation and explanatory power.

3.2.2 | The feature extraction scheme based on spectral Euclidean distance

EDFS is a technique used for analysing the spectral characteristics of signals. The fundamental idea of this scheme is to decompose the signal into spectral segments and calculate the Euclidean distance between these segments to extract a feature vector that reflects the spectral features of the signal.

A rapid mode recognition scheme based on spectral analysis for the dynamic sensing domain was proposed in 2015 [18]. After obtaining spectral features, the scheme calculates the Euclidean distance between the signal spectrum and the template to recognise and classify disturbance signals. Figure 4a shows the temporal signals with three types of disturbances. Figure 4b displays the results after applying short-time energy and short-time shift difference processing to the signals. Figure 5a shows the normalised results of the three types of disturbance signals and natural wind signals, while Figure 5b presents the spectra of these four typical signals.

Experimental results indicated that this scheme can effectively identify disturbance signals. Compared with traditional

dynamic time–domain pattern recognition schemes, it has a faster processing speed, requires fewer training samples in the experiments, and exhibits noise suppression ability.

EDFS allows for an intuitive comparison of the similarity and dissimilarity between signal spectra, has been found suitable for analysing signals such as sound signals and vibration signals that possess spectral characteristics. However, in practical applications, it places higher demands on preprocessing to eliminate the impact of varying signal lengths and amplitude scales.

3.2.3 | The feature extraction scheme based on waveform characteristics

The waveform characteristics is a scheme that utilises the time–domain waveform of a signal for feature extraction. It has been found suitable for analysing various waveform features, such as amplitude, duration, and slope to capture the temporal characteristics of the signal.

A scheme for distinguishing and recognising interference by extracting time–domain and frequency–domain features, such as mean, variance, root mean square, and signal power from signals, was proposed in 2017 [26]. A scheme for using underground fibre optics for detecting and characterising thunder-induced ground motions was proposed in 2019, which

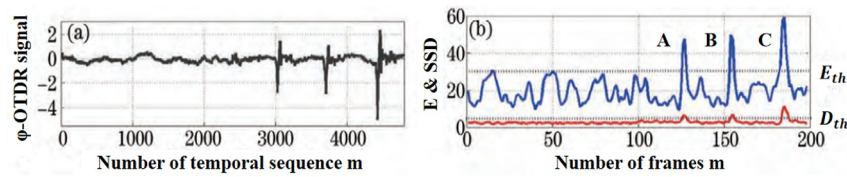


FIGURE 4 (a) ϕ -OTDR signals in temporal sequence at a certain position with three disturbances; (b) signal energy (blue) and SSD (red) versus frame number with their threshold E_{th} and D_{th} [18]. SSD, Short-Time Shifted Difference.

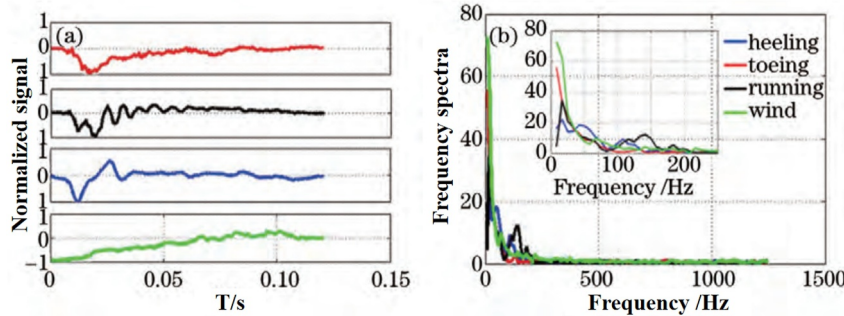


FIGURE 5 (a) Typical signals of heeling (simulating climbing a wall), toeing, running and natural wind; (b) their frequency spectrum [18].

uses arrival times of the 18 events to estimate the phase velocity of the near surface, the back azimuth, and location of thunder-seismic sources that are verified with lightning locations from the National Lightning Detection Network [27]. A scheme for temperature monitoring of large-scale infrastructure through image denoising of raw edge node data in facility monitoring networks was proposed in 2021 [28].

A scheme aimed at enhancing the safety of underground pipelines was proposed in 2022, and a comparative analysis of four typical vibration signals of buried pipelines was conducted [19]. The three types of features were selected for feature extraction from the aforementioned samples. The feature comparison of different disturbance events is shown in Figure 6.

From the Figure 6, significant differences in the features of different disturbance signals can be observed. After subsequent pattern recognition, the system false alarm rate was reduced by 99.6%, resulting in a final recognition rate of 94.8%.

The waveform characteristics has been found suitable for various types of signals, such as audio signals, vibration signals, and biomedical signals, which exhibit temporal waveform characteristics. However, this scheme is sensitive to noise and is limited by susceptibility to noise disturbance and signal shape scale constraints. Therefore, it needs to be adjusted according to the specific signal type.

3.2.4 | The feature extraction scheme based on Fourier transform

The feature extraction scheme based on Fourier transform is a technique that utilises the spectral information of a signal for feature extraction. Several steps are involved, including signal

preprocessing, Fourier transform, spectral analysis, and feature extraction. With advancements in computer and mathematical technologies, Fourier transform has undergone rapid development in the fields of signal processing and feature extraction.

In the process of employing Fourier transform, the recognition rate can also be affected by the window function and window length. A scheme using a Kaiser window with a length of 4800 for performing short-time Fourier transform (STFT) on intrusion signals was proposed in 2019 [20]. This article had used a Hanning window with a length of 4800 and a Kaiser window with a side lobe of 0.5 to perform STFT on four types of intrusion signals. The results obtained are shown in Figure 7.

From the Figure 7, it can be seen that compared with the Hanning window, the Kaiser window retains more main lobe energy and time–frequency information, reduces side lobe amplitude, eliminates high-frequency noise interference, and better reflects the true spectrum of the original intrusion signal.

A scheme that addresses the problem of high sensitivity and difficulty in distinguishing target events from background signals caused by interference was proposed in 2020. The scheme involved the analysis of high-speed train motion characteristics and the extraction of signal features from both the time and frequency domains. A time–frequency domain feature set was constructed to extract signal characteristics [21].

The Fourier transform has been found suitable for periodic signals and signals with prominent frequency spectrum characteristics, such as audio signals and image signals. Additionally, the Fourier transform has complementary properties to other signal processing schemes, enabling its combination with time–domain and wavelet–domain analyses, among others, to extract more abundant and diversified features.

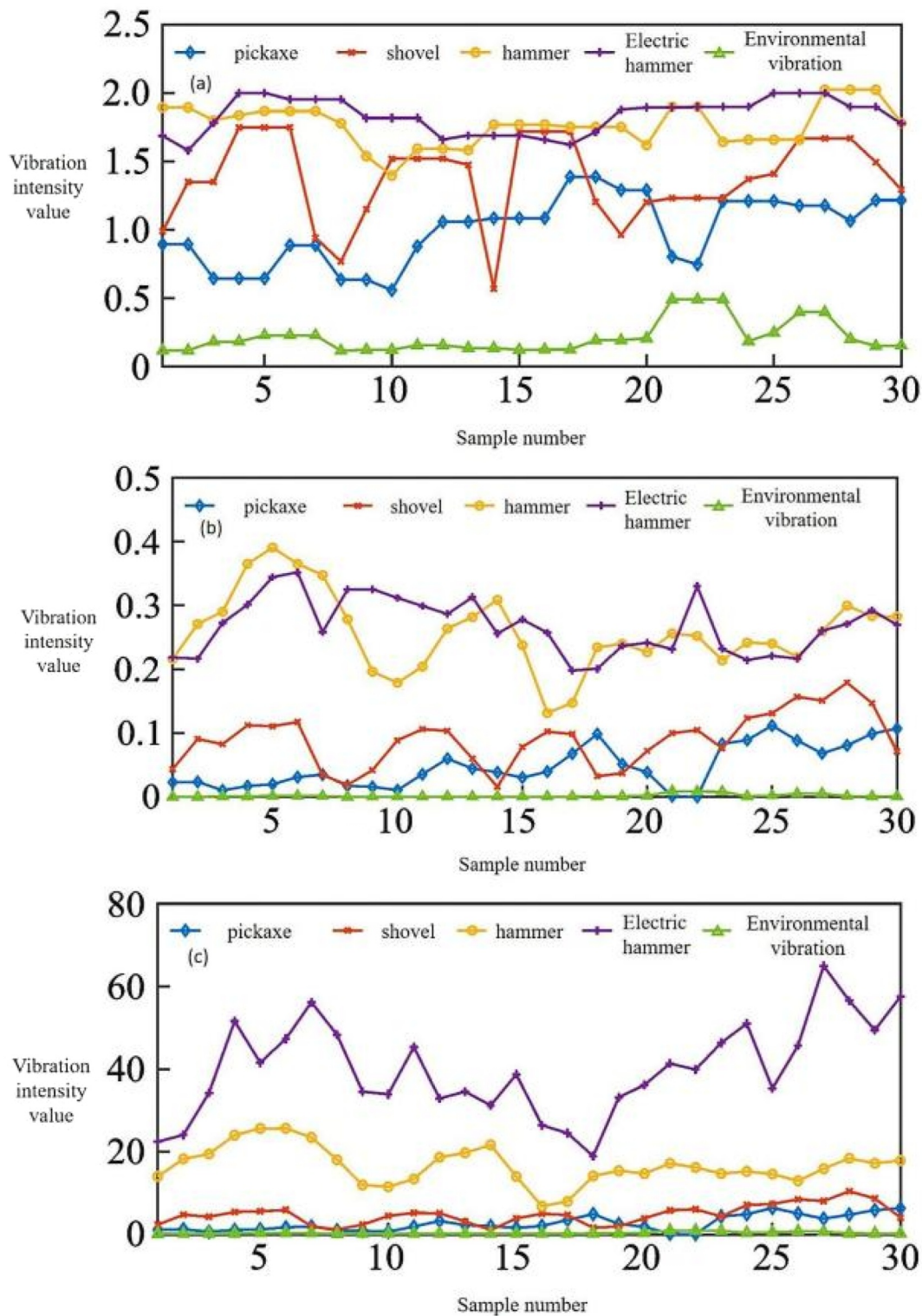


FIGURE 6 Feature comparison of different disturbance signals (a) maximum value feature comparison; (b) variance feature comparison; (c) short-time energy feature comparison [19].

3.2.5 | The feature extraction scheme based on empirical mode decomposition

EMD is a signal decomposition scheme based on adaptive local features, used to extract the intrinsic modes of a signal. It was

proposed in 1998 [29]. This scheme decomposes the signal into a series of intrinsic mode functions (IMF), where each IMF represents a local oscillatory mode within the signal. Signal features can be extracted from the decomposed IMF, such as computing the mean, variance, energy, and frequency of each IMF.

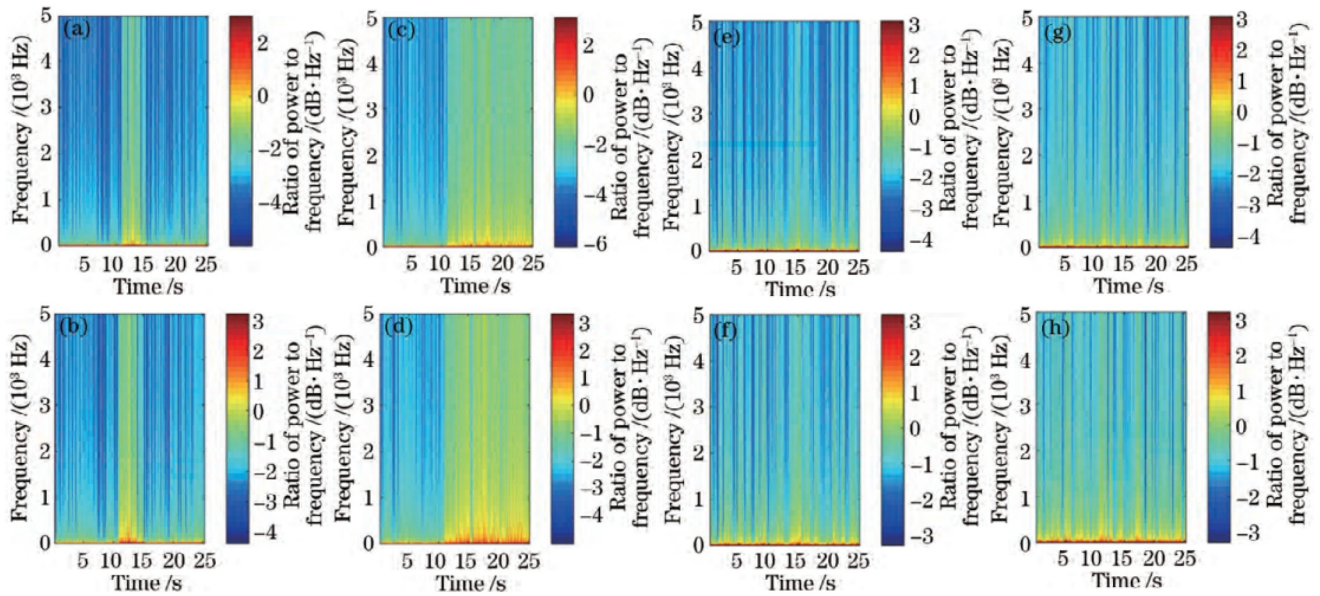


FIGURE 7 STFT time–frequency diagrams of two kinds of window functions for processing four intrusion events. (a), (c), (e), (g) Time–frequency diagrams of knocking, shaking, winding, and raining signals after passing through the Hanning window; (b), (d), (f), (h) time–frequency diagrams of knocking, shaking, winding, and raining signals after passing through the Kaiser window [20]. STFT, short-time Fourier transform.

In order to improve the accuracy of intrusion recognition, a recognition scheme combining EMD with 1-D CNN was proposed in 2021. This improved scheme has effectively identified intrusion vibration signals collected in environments [30]. In the same year, an improved EMD scheme for DVS systems was proposed [31].

A feature extraction scheme that combines information entropy-optimised EMD with improved Mel-frequency cepstral coefficients was proposed in 2023 to enhance the identification accuracy of compressed gas leak signals [22].

EMD has been found suitable for analysing non-linear and non-stationary signals, including vibration and seismic signals, effectively addressing intricate time–frequency characteristics. It has been recognised for fully extracting underlying components, enhancing feature discriminability, and addressing spectral aliasing problems in these types of signals. Its versatility across signal types and capacity to capture intrinsic characteristics have led to its extensive application in signal processing and feature extraction tasks. Notably, EMD excels in efficiently handling complex signal components and adapting to diverse patterns without necessitating predefined assumptions.

3.2.6 | Feature extraction schemes based on linear discriminant analysis

Feature extraction schemes based on LDA are classical techniques used for dimensionality reduction and classification. LDA aims to maximise the between-class distance and minimise the within-class distance in order to project the original high-dimensional data into a lower-dimensional space, thus extracting discriminative features.

A supervised dimensionality reduction technique was proposed in 2000 to address pattern recognition and classification

problems [32]. Subsequently, LDA was applied in various domains such as face recognition, text classification, and biomedical signal processing. Specifically, the LDA scheme includes the following steps: (1) the computation of the within-class scatter matrix and between-class scatter matrix. (2) The selection of the projection direction. (3) The performance of dimensionality reduction and feature extraction.

A supervised dimensionality reduction scheme using LDA was employed in 2018 to address the classification and localisation problems of interference signals [23]. The scheme involves extracting feature vectors from the time–domain signals of three types of disturbance events, and the final results have demonstrated that this approach achieves good recognition performance.

LDA has been found suitable for supervised learning and multiclass problems, and it has the capability to extract discriminative features. In cases where the sample sizes of different classes in the dataset are imbalanced, the LDA scheme may favour the class with more samples.

4 | PATTERN RECOGNITION SCHEMES

In the recognition of fibre optic intrusion events, both unsupervised and supervised learning methods are typically involved. Unsupervised learning refers to modelling and analysing patterns from samples without artificial labels, such as autoencoder and other unsupervised learning methods. An unsupervised anomaly detection method was successfully applied to DAS systems, effectively detecting intrusion events within a distance of 35.0 km [33].

In contrast, supervised learning involves training models using labelled training samples and applying them to unknown

samples to obtain the best output results. Supervised learning methods include traditional feature engineering and statistical models, as well as neural network classifiers based on deep learning. Traditional methods rely on feature extraction and classification algorithms to identify patterns in optical signals, while neural network methods utilise deep learning models to learn and identify optical signals, demonstrating strong adaptive and non-linear modelling capabilities.

4.1 | Traditional recognition classifier

The classifier is a crucial component in intrusion event recognition, as it classifies different types of intrusion events based on the data's features. Some of the classical schemes include support vector machine (SVM), k-nearest neighbor classifiers, random forest (RF) etc. This paper mainly focuses on introducing SVM and RF. The specific models are shown in Table 3.

4.1.1 | Support vector machine

SVM is a widely used scheme in the field of pattern recognition, known for its efficient learning capabilities, strong noise resistance, and low structural risk. This algorithm is mainly employed to address binary classification problems. For a dataset containing two classes of events, SVM aims to find a hyperplane that maximally separates these two classes, ensuring that events of the same class are grouped together on the same side of the hyperplane.

The SVM algorithm has originated from the statistical learning theory proposed in the late 1960s [41] and has since undergone further development and promotion. Subsequently, the SVM algorithm has been applied in various fields, such as pattern recognition, image recognition, and text classification [42].

An SVM scheme was proposed in 2012 to classify optical fibre signals based on the energy distribution characteristics of

the signals [43]. An SVM scheme was proposed in 2014 to determine intrusion location and type from optical fibre vibration signals [44]. An SVM scheme with a left-skewed binary tree structure was proposed in 2017 to achieve a correct recognition rate of 96.0% for disturbances [45].

In response to disturbances occurring on the fence, a scheme utilising SVM for classification and recognition was proposed in 2018. Experimental results have demonstrated an overall recognition rate of 98.1% for the four types of signals [34].

A scheme enhancing the classification capability of a φ -OTDR system for multiple similar events was proposed in 2020. This scheme combines deep learning with traditional classifiers, and the system architecture for SVM training using features extracted from a CNN model is illustrated in Figure 8 [35].

A Φ -OTDR system based on intensity was established in 2023, which employs an SVM to recognise six types of events, achieving an average recognition rate of 82% [36].

A scheme based on Laplacian feature selection, fused with AdaBoost and M1-K-Nearest Neighbors (KNN), was proposed in 2021. The recognition accuracy of this scheme was reported as 95.4% [37]. Concurrently, in the same year of June, a scheme principal component analysis network in conjunction with SVM was proposed. Experimental results demonstrated a recognition accuracy of 99.3% for three types of signals [38]. For monitoring of water supply pipelines, a two-stage SVM scheme was proposed in 2022 [39].

SVM has been found suitable for binary classification problem. It offers advantages such as high-dimensional feature space classification, robustness, and strong generalisation capability. However, it has some limitations, including sensitivity to parameter selection (e.g. C parameter, kernel function parameter) and high computational complexity. When dealing with large-scale datasets or cases involving multiple classes, SVM may require longer training time and larger storage space.

TABLE 3 Traditional recognition classifiers.

Pattern recognition scheme	Ref.	Specific identification event	Average recognition accuracy (%)
Support vector machine (SVM)	[34]	Blowing, raining, knocking, cutting	98.1
	[35]	Background noise, walking, jumping, knocking, excavation, cycling, raining, and watering	94.1
	[36]	Six types of events	82.0
	[37]	Shaking, climbing, and cutting	95.4
	[38]	Hitting, clapping on the net, and shaking	99.3
	[39]	Manual excavation, human contact, pipeline leakage, water flow in pipeline cavity, and mechanical excavation	95.4
Random forest classifier (RF)	[40]	Pressing, watering, and tapping	96.5

Abbreviations: RF, random forest classifier; SVM, support vector machine.

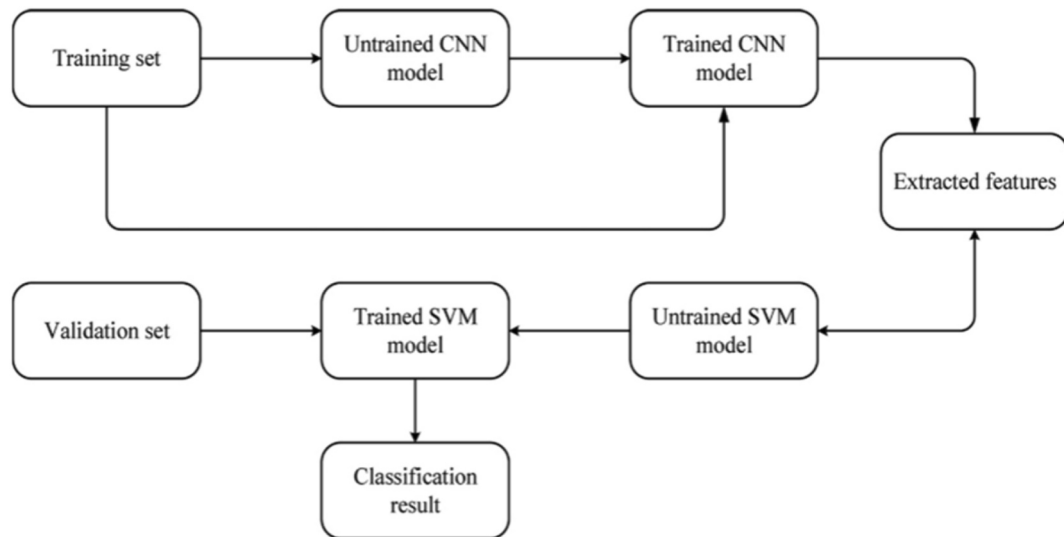


FIGURE 8 Structural diagram for extracting features from the CNN model for SVM training [35]. CNN, convolutional neural network; SVM, support vector machine.

4.1.2 | Random forest classifier

RF is an ensemble learning algorithm. It constructs multiple decision trees by randomly selecting feature subsets and sample subsets. The final decision is made by voting or averaging the prediction results from these trees. The basic ideas of the RF algorithm include random sampling, decision tree construction, and prediction decisions [46].

The RF algorithm was originally proposed in 2001 [47] and has since been further improved and popularised, for example, the quantile regression forest [48] and the random survival forest [49]. The forest algorithm has been applied in various fields such as pattern recognition, image processing, and text classification [50].

A disturbance event recognition scheme based on a RF classifier was proposed in 2019 for reducing the false alarm rate of φ -OTDR systems [40]. Experimental results indicated that the recognition rates for four types of signals were 97.0%, 93.7%, 97.3%, and 98.1%.

RF algorithm has been found suitable for handling multi-class problems, high-dimensional data, and large-scale datasets, offering advantages such as high accuracy, interpretability, and robustness. However, it requires significant computational resources and storage space during the construction of multiple decision trees and making predictions. Additionally, some parameters need to be adjusted, such as the number of decision trees and the size of feature subsets.

4.2 | Neural network recognition classifier based on deep learning

The neural network recognition classifier is an advanced pattern recognition technique based on deep learning schemes. In contrast to traditional recognition classifier, it effectively emulates the functioning of the human visual system, allowing

them to learn intricate feature representations of complex data types such as images and videos. They possess enhanced non-linear modelling capabilities and adaptive learning capabilities. By combining and connecting multiple layers of neurons, high-level abstract features are automatically extracted from raw data, enabling the learning and recognition of complex patterns and regularities.

Neural networks have gone through a process of evolution, starting from the early perceptron model, then the rise of multi-layer feedforward networks and SVM, and finally to the emergence and explosion of deep learning. With continuous improvements in algorithms and computational resources, neural networks have achieved numerous breakthroughs in pattern recognition, computer vision, and other fields. Among the deep learning-based neural network recognition classifiers are CNN, backpropagation neural network (BPNN), probabilistic neural networks, long short-term memory neural networks (LSTMNN) etc. This paper mainly focuses on CNN and BPNN. The specific models are shown in Table 4.

4.2.1 | Convolutional neural network

CNN is a deep-learning algorithm used for pattern recognition tasks in the field of computer vision. Its core components include convolutional layers, pooling layers, and fully connected layers. Through large-scale training data and appropriate optimisation algorithms, CNN can automatically learn higher-level feature representations, thereby improving the accuracy of pattern recognition.

The artificial neural neuron model was proposed in 1943, abstracting the working principles of biological neurons into mathematical models and laying the foundation for neural networks [56]. A scheme introducing the perceptron learning algorithm, incorporating weights and thresholds, was proposed in the 1950s and 1960s, used for binary classification tasks [57].

TABLE 4 Deep learning-based neural network recognition classifiers.

Pattern recognition scheme	Ref.	Specific identification event	Average recognition accuracy (%)
Convolutional neural network (CNN)	[51]	Detection of reflection events and non-reflection events	99.0
	[52]	Manual excavation, mechanical excavation	97.0
Backpropagation neural network (BPNN)	[53]	Vehicle passage, train passage, manual tapping, and mechanical construction	81.4
	[54]	Anchor impact, anchor vibration, seismic waves, and ship wake	88.0
	[55]	Hammer impact, rope dragging, and no intrusion	100.0

Abbreviations: BPNN, backpropagation neural network; CNN, convolutional neural network.

The Back Propagation (BP) algorithm was proposed in the 1980s, enabling the training of multi-layer feed-forward neural networks [58]. Deep belief networks and CNN have been proposed in the 2000s and widely applied in fields such as image and speech processing. With the widespread availability of large datasets and computing resources, deep learning was applied since 2010 in computer vision, natural language processing, speech recognition, and more [59]. Currently, the development of neural networks continues, with researchers constantly proposing new network structures and training algorithms, driving their application and further advancement across various domains.

A wavelet threshold denoising scheme based on grasshopper optimisation was proposed in 2022 for rapid identification of anomalies in long-distance optical fibres [51]. This scheme optimises the parameters in the wavelet threshold denoising using the grasshopper optimisation algorithm. Finally, the identified results are further recognised using the CNN, resulting in an improvement of 6.4% in the recognition rate of reflected events and 18.9% in the recognition rate of non-reflected events.

The application of integrated neural networks has enabled more effective identification of intrusion signals. A scheme that combines CNN, LSTMNN, and deep neural networks into a unified architecture called convolutional-LSTM-DNN (CLDNN) was proposed in 2019 [52]. This approach was applied to address safety monitoring problems in oil and gas pipelines. The architecture of the CLDNN neural network is illustrated in Figure 9. Experimental results indicated that this scheme has achieved an average recognition rate of 97.0%.

A combined neural network of LSTM and CNN was proposed in 2022 [60], and its network architecture is depicted in Figure 10. Experimental data shows that the LSTM-CNN scheme achieved a test set accuracy of 94.6% and a training set accuracy of 97.8%.

CNN have the capability to automatically learn and extract local features from images, achieving high classification accuracy. Moreover, target object detection and localisation in images can be achieved through feature map analysis and classification using CNN. However, neural network training demands large amounts of data and often consumes considerable time. In practical applications, certain parameters of the neural network, such as network architecture and iteration count, need to be adjusted to achieve better performance and accuracy.

4.2.2 | Backpropagation neural network

BPNN is a common type of feedforward neural network. It consists of three parts: the input layer, the hidden layer, and the output layer. By establishing connections between multiple layers of neurons and adjusting their weights, BPNN can learn and classify complex problems [53].

A scheme incorporating a fusion classification algorithm based on BP and SVM was proposed in 2021 [54], and this scheme was applied to address pipeline security problems. The input feature vectors are separately fed with the same feature value data into two distinct classifiers for independent training and pattern recognition. The corresponding vibration event will be identified only when the results from these two classifiers are consistent. The architecture of the fused classifier is illustrated in Figure 11. Experimental results indicated that the fusion classifier algorithm based on BP and SVM has achieved an average recognition rate of 81.4%.

A scheme utilising BP neural network for practical disturbance classification and identification of underwater optical cables was proposed in 2022 [55]. Experimental results indicated that this scheme achieved a recognition accuracy of 88.0% for underwater cable disturbance signals.

A scheme using BP neural network for underwater cable anchor rod damage event detection was proposed in 2023 [61]. Experimental results indicated that the recognition accuracy of this scheme is 100.0%.

BPNN is a classical neural network model that has been found suitable for tasks such as pattern classification, function approximation, prediction, and regression. In practical applications, the model's performance can be enhanced by carefully designing the network architecture, selecting appropriate parameters (e.g. the number of neurons in the hidden layers), and using optimisation algorithms (e.g. stochastic gradient descent).

5 | CONCLUSIONS AND PERSPECTIVES

In the process of information source identification for DAS, the complexity of monitoring environments often results in false alarms and missed alarms. Therefore, the accurate detection and differentiation of disturbance events and their types in the presence of external noise disturbance are crucial

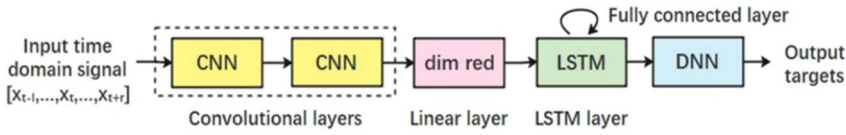


FIGURE 9 CLDNN architecture [52]. CLDNN, convolutional-LSTM-DNN.

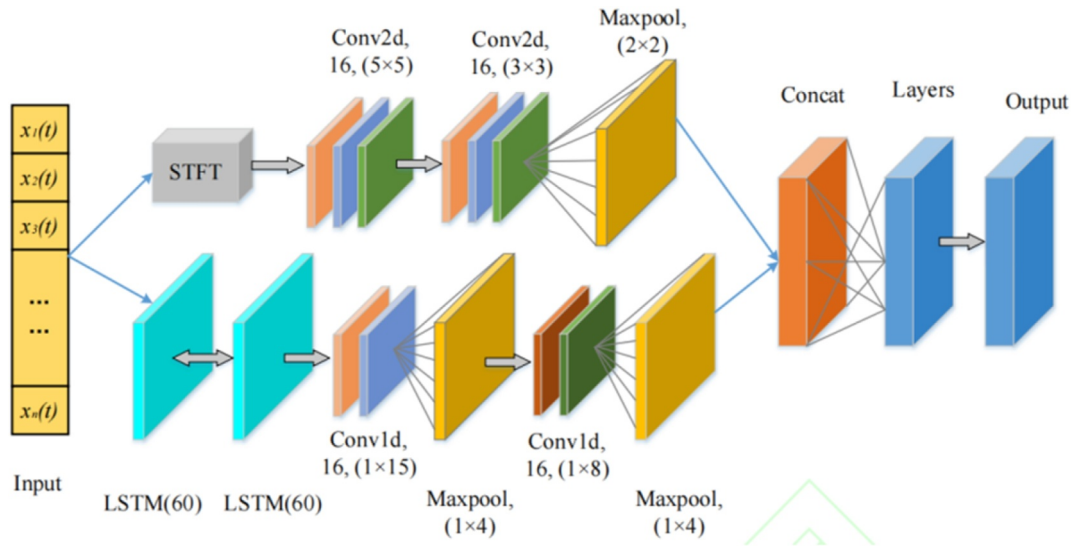


FIGURE 10 LSTM-CNN network architecture [60].

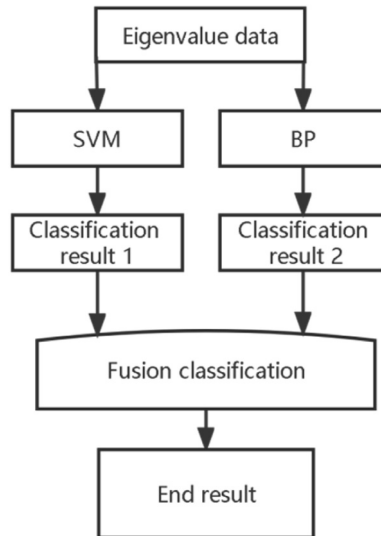


FIGURE 11 Schematic diagram of the fusion classifier algorithm [54].

in practical applications. This paper provides an overview and comparison of feature extraction and pattern recognition schemes for identifying different events using DAS and specific applications. Detailed results are presented in Table 5.

DAS technology has potentially encountered more complex application scenarios. To address these intricate contexts, further optimisation of feature extraction and pattern recognition techniques is needed. From a technical perspective, emphasis has been placed on enhancing the system's data processing capabilities and ensuring the reliability of the

constructed models. Simultaneously, continuous iteration and improvement of technology and algorithms have been deemed necessary to comprehensively optimise system performance. Performance and robustness have been enhanced through parameter adjustments and the selection of model structures. A trend has emerged towards the integration of multiple features in feature extraction, aimed at enhancing the accuracy and robustness of pattern recognition. By combining different feature types, such as time–frequency, spatial, and spectral information, a more comprehensive data representation has been captured, facilitating the efficient processing of complex signals. Furthermore, compared to utilising individual classifiers or neural networks separately, the use of integrated neural networks or combining traditional classifiers with neural networks has generally yielded superior recognition accuracy. It has been shown that this approach has the advantage of harnessing the strengths of diverse technologies, thus achieving enhanced performance in more practical applications.

The wavelet characteristics has been found suitable for handling non-stationary signals and signals with significant frequency variations. Different wavelet basis functions can be utilised for decomposition to capture more comprehensive time–frequency information. EDFs has been found suitable for analysing signals with spectral characteristics such as sound and vibration signals. To enhance its performance, advanced spectral analysis schemes such as higher-order spectral analysis can be introduced to better capture the signal's frequency domain features. The waveform characteristics has been found suitable for various types of signals with temporal waveform features. More sophisticated waveform analysis techniques

TABLE 5 Comparison of different approaches in literature.

Ref.	Feature extraction scheme	Pattern recognition scheme	Average recognition accuracy (%)
[14]	Wavelet packet decomposition	SVM	97.0
[15]	Peak-to-mean ratio, segment length, average segment interval, and wavelet packet decomposition	SVM	98.3
[16]	Wigner–Ville distribution	Gradient boosting algorithm	93.3
[19]	Compute maximum value, variance, and short-time energy	RF	94.8
[20]	Short-time Fourier transform	CNN	97.2
[21]	Fast Fourier transform	SVM	
[22]	Improved empirical mode decomposition combined with improved Mel frequency cepstral coefficients (MFCC)	SVM	94.7
[23]	Linear discriminant analysis	KNN	92.6
[34]	Meir frequency cepstrum coefficient	SVM	98.1
[35]	CNN	SVM	94.1
[36]	Multi-characteristic parameter	SVM	82.0
[38]	Principal component analysis network	SVM	99.3
[39]	Equilibrium optimisation algorithm, variational mode decomposition, and singular spectrum analysis	SVM	95.4
[40]	Random forest classifier	RF	96.5
[51]	Two-point scheme, least square scheme, and wavelet reconstruction based on locust optimisation	CNN	99.0
[52]	Time domain signal filter cutting	CNN + LSTM + DNN	97.0
[54]	The ratio of points over and not over the threshold, the signal width, the short-time wavelet packet coefficient, and the short-time energy average	BP + SVM	81.4
[55]	Disturbance duration and variance and energy distribution	BPNN	88.0
[61]	Short-time energy, time–frequency entropy, short-time threshold rate, centre of gravity frequency	BPNN	100.0

Abbreviations: BPNN, backpropagation neural network; CNN, convolutional neural network; KNN, K-Nearest Neighbors; MFCC, mel frequency cepstral coefficients; RF, random forest classifier; SVM, support vector machine.

such as transient analysis or periodicity analysis can be introduced to extract more specific waveform features. The Fourier transform has been found suitable for periodic signals and signals with prominent spectral features. Preprocessing using window functions can be applied to reduce leakage effects and enhance the accuracy of spectral analysis. EMD has been found suitable for non-linear non-stationary signals and signals with complex time–frequency features. Different decomposition algorithms, such as variational mode decomposition, can be tried to adapt to different signal types. LDA has been found suitable for supervised learning and multi-class problems. Combination with other dimensionality reduction schemes such as Principal Component Analysis has been found suitable for obtaining more discriminative features.

SVM algorithm has been found suitable for binary classification problems, applicable to both linearly separable and non-linearly separable cases, and its performance can be enhanced through careful kernel selection, parameter tuning, and the utilisation of advanced kernel functions. RF algorithm

has been found suitable for handling multi-class problems, high-dimensional data, and large-scale datasets and can be improved through techniques such as feature importance analysis and ensemble pruning. CNN algorithm has been found suitable for processing spatially correlated signals with a large number of samples, and their performance can be optimised through techniques such as transfer learning, parameter tuning and data augmentation. BPNN algorithm has been found suitable for tasks such as pattern classification, function approximation, prediction, and regression, and their effectiveness can be elevated through architecture optimisation, activation function selection, and regularisation techniques.

Through case studies, it has been observed that under the same classifier, feature extraction algorithms involving multiple feature fusion often achieve higher recognition accuracy. For instance, in refs. 16, 26, and 42, the adoption of multi-feature fusion for feature extraction has resulted in average recognition rates of 98.3%, 94.8%, and 90.0%, respectively. Simultaneously, a comparison between refs. 16 and 26 indicates that the

utilisation of comprehensive time–frequency domain features yields better recognition accuracy compared to using only time–domain features.

It has been demonstrated in ref. [11] that adopting the LSTM-CNN scheme is superior to using the CNN scheme and Artificial Neural Network scheme. When faced with validation data beyond the training samples, the trained LSTM-CNN outperforms four common classifiers, namely decision trees, RF algorithms, SVM, and KNN. Furthermore, in ref. [59], the adoption of the fusion classification algorithm based on BP and SVM has shown significantly higher recognition accuracy than individual classifier algorithms.

In summary, the continuous development of DAS technology has highlighted the crucial role of multi-feature fusion feature extraction schemes and ensemble classifier techniques, which have led to significant improvements in the system's recognition accuracy. Moving forward, the emphasis should be on further enhancing feature extraction and pattern recognition techniques, exploring additional fusion algorithms, and integrating the latest artificial intelligence and machine-learning technologies to enhance the reliability of feature extraction and pattern recognition systems. This will offer superior solutions to monitoring and recognition challenges in practical application scenarios, thereby promoting the widespread application of DAS technology in various fields. Through continuous exploration and innovation, we can expect to achieve breakthroughs and advancements in DAS technology, enabling higher efficiency and reliable monitoring and recognition capabilities for various industries.

AUTHOR CONTRIBUTIONS

Yage Zhan: Funding acquisition; project administration; resources; software; supervision. **Lirui Liu:** Conceptualization; investigation; software; visualization; writing – original draft; writing – review & editing. **Kehan Li:** Data curation; formal analysis.

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CONFLICT OF INTEREST STATEMENT

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

DATA AVAILABILITY STATEMENT

The data used to support the findings of this study are available from the corresponding author upon request.

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